CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in agentic AI systems that automate business workflows, from multi-agent LLM orchestration and RAG to validation against ground truth, cost benchmarking and cloud deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by consulting work with business stakeholders at Covestro and a strong foundation in machine learning and Python.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Designed and optimized a multi-agent system of LangGraph AI agents (entity extraction, relationship analysis, consolidation and reflection, running in parallel) for knowledge extraction on complex regulatory and technical documents, validated against an SAP-system-derived ground truth; precision/recall raised from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Sole developer owning the full knowledge-extraction pipeline end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph to agents through an API with millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a human-in-the-loop graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts validate and approve knowledge-graph changes.
- Used Claude Code, custom skills and agentic coding workflows to prototype fast and ship reliable code grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, MCP, Neo4j, MongoDB, Postgres, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows and translated them into an architecture; designed and deployed an agentic LangGraph application that routes each request to vector database queries, general inference or summarization.
- Developed a Python backend with FastAPI APIs and a React frontend, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, React, Postman**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built event-driven AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.

- Benchmarked multilingual embedding models on over 1M chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, Vector Databases, Azure AI Search, AWS (Lambda, SQS, DynamoDB, CDK), Docker, Streamlit**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) for the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time series analysis of EV load.
- Developed an LSTM that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Agent Eval Harness | LangGraph | Guardrails | OpenTelemetry | GitHub Actions

- Tool-calling LangGraph agent for sea-freight logistics booking requests over mocked TMS/ERP APIs, with guardrails, a validation layer for every tool call, fallback to a rules path and human-in-the-loop approval above a threshold; a golden-set harness in CI benchmarks accuracy, latency and cost, with logging and monitoring via OpenTelemetry.

TECHNICAL SKILLSET

- **Agentic AI:** prompt engineering, LangGraph, LangChain, multi-agent orchestration, LLM APIs (LiteLLM), RAG, GraphRAG, model tuning (QLoRA, retrieval fine-tuning), MCP
- **Reliability:** validation against ground truth, deterministic (code-based) extraction pipelines, human-in-the-loop review, evaluation (RAGAS, PytrecEval), cost and latency benchmarking
- **Engineering:** Python, FastAPI, REST APIs, Python microservices, React (beginner), Postman, Claude Code, Git, GitLab CI/CD, Agile
- **Data & Cloud:** SQL/NoSQL (Postgres, MongoDB, Neo4j), vector databases (Qdrant, Pinecone, Azure AI Search), AWS (Lambda, SQS, DynamoDB, EC2, S3, CDK), Azure, Docker, Kubernetes
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles